



Painting in a Painting - AI for Hidden Image Detection in Artworks

Submitted to: Google Summer of Code 2025
Under: ArtExtract Initiative

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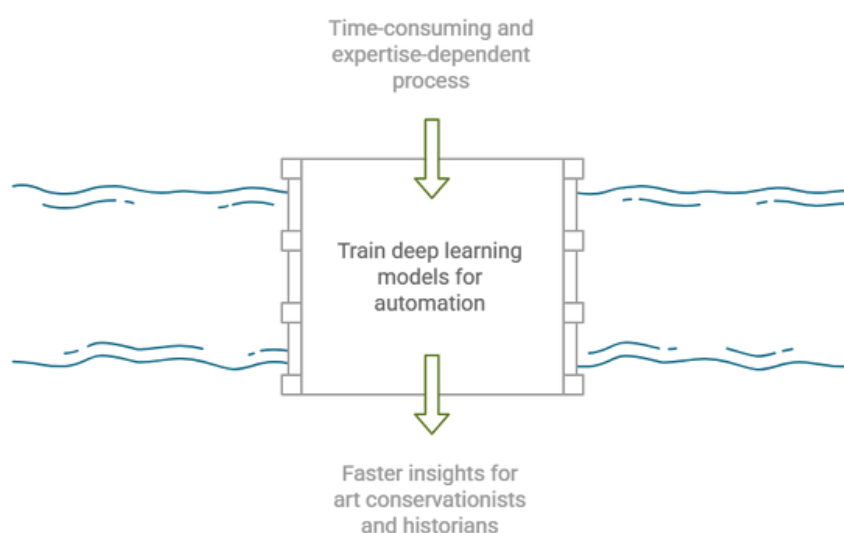
Timezone: IST (GMT + 0530)

"The layers of a painting are like layers of history — I want to use machine learning to unearth both."

Synopsis

Art restoration and historical analysis have been significantly enhanced by the use of multi-spectral imaging and artificial intelligence, enabling experts to uncover hidden layers beneath paintings. However, manual interpretation of such data is time-consuming and relies heavily on domain expertise.

This project aims to automate the process of hidden image detection by training deep learning models on multi-spectral painting datasets. As part of the ArtExtract initiative, the outcome will support art conservationists, historians, and researchers in analyzing pigment composition, restoration history, and underlying artworks more efficiently.



Benefits to the Community

- Automated Art Analysis: Reduces the reliance on manual inspection by enabling AI-assisted detection of hidden artworks and painting modifications.
- Restoration Insights: Assists conservationists in identifying previous restorations, pigment degradation, and canvas reuse, improving restoration planning.
- Historical Discovery: Provides art historians with tools to uncover lost compositions and concealed artistic elements, contributing to cultural heritage preservation.
- Open-Source Contribution: Delivers a scalable and well-documented AI pipeline that can serve as a foundation for future research and practical applications in art and AI.

Personal Introduction:

I am a Computer Science undergraduate with a strong interest in the intersection of artificial intelligence and art. My recent work has focused on computer vision, deep learning, and large-scale data processing—especially through tasks related to artwork classification and retrieval. These experiences have equipped me with both technical proficiency and a deep appreciation for art-driven AI applications.

My current academic semester will conclude by late April, giving me ample time to prepare and transition into the GSoC project.

I will be able to dedicate approximately 35 hours per week to the project, in line with GSoC's expectations for medium-length projects. I am also open to putting in additional time when necessary to ensure timely progress and high-quality outcomes. While my new semester begins in August, I will plan my schedule to maintain consistent progress, with a buffer period already accounted for toward the final weeks of the coding phase. I am committed to giving my full effort and am always open to going the extra mile if the work requires it.



Test Task Implementation

For the test tasks, I worked on two different yet complementary problems:

- (1) a convolutional-recurrent model for art classification**
- (2) a similarity model for finding related artworks.**

Both tasks required building scalable, efficient pipelines for handling large datasets and implementing robust evaluation methods to ensure model performance and reproducibility.

Task 1: Convolutional-Recurrent Model for Art Classification



Objective:

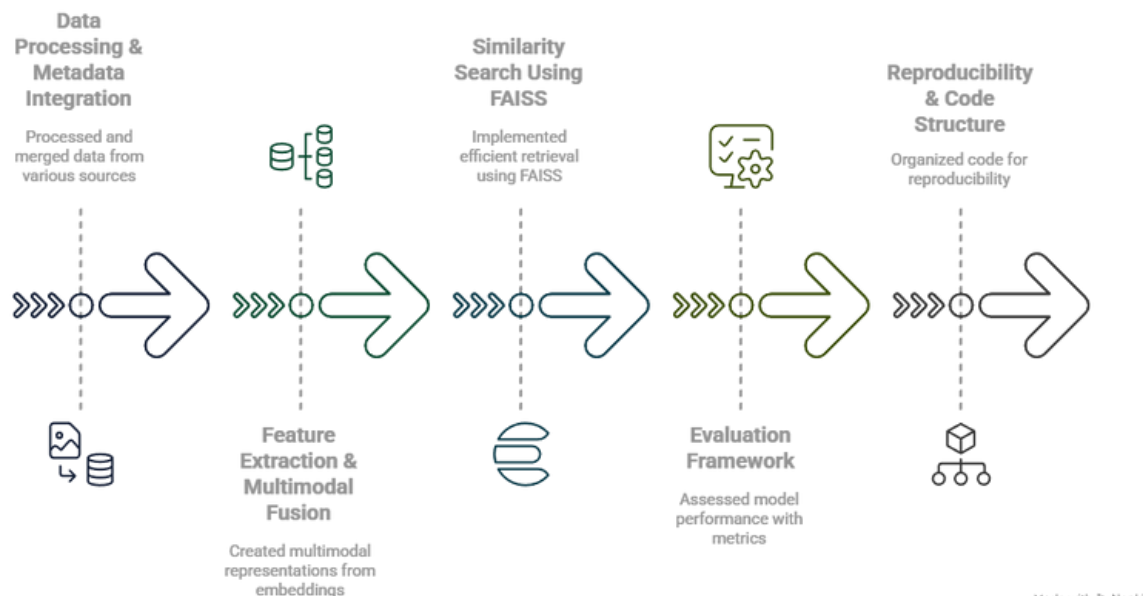
Develop a model to classify paintings based on Style, Artist, and Genre using convolutional-recurrent architectures. The goal was to build a robust framework that could handle the complexities of artistic datasets while ensuring generalization and interpretability.

Implementation Workflow:

- Data Preparation
 - Processed the WikiArt dataset, focusing on three attributes: Artists (23 classes), Genres (10 classes), and Styles (27 classes).
 - Resized images to 224×224, applied normalization, and stored batches using NPZ compression to handle 25GB+ of image data efficiently.
 - Implemented validation splits per attribute to ensure fair performance evaluation.

- Model Development
 - Used a ResNet50-based feature extractor for visual embeddings.
 - Applied Global Average Pooling to reduce dimensionality, followed by Batch Normalization and Dropout (0.75) for regularization.
 - Introduced L2 regularization ($\lambda=0.1$) to minimize overfitting.
 - Implemented adaptive learning rate reduction using ReduceLROnPlateau for stable convergence.
- Evaluation & Outlier Detection
 - Used Precision, Recall, F1-Score, and Confusion Matrices for classification performance.
 - Implemented t-SNE visualizations to analyze embedding distributions.
 - Applied multiple outlier detection methods:
 - Centroid distance thresholding (95th percentile cutoff).
 - K-Means clustering anomalies.
 - Isolation Forest with contamination rate of 5%.

Task 2: Similarity Model for Paintings



Objective:

Develop a multimodal similarity search system to find related artworks based on visual and textual metadata. The challenge was to build a scalable search solution that integrates deep learning embeddings and efficient retrieval mechanisms.

Implementation Workflow:

- Data Processing & Metadata Integration
 - Processed 100K+ images from the National Gallery of Art dataset, resizing them to 224×224 and normalizing with ImageNet statistics.
 - Merged 5 metadata sources (titles, artist info, subject taxonomy, etc.) into a unified dataset.
 - Used chunked processing for handling large-scale metadata efficiently.
- Feature Extraction & Multimodal Fusion
 - Used CLIP (ViT-B/32) for 512D visual embeddings and SentenceTransformer (all-MiniLM-L6-v2) for 384D text embeddings.
 - Combined them into an 896D multimodal representation with a weighted ratio of 70% visual and 30% textual features.

- Similarity Search Using FAISS
 - Implemented FAISS-based nearest neighbor search for fast retrieval.
 - Used L2-normalized vectors with cosine similarity for accurate comparisons.
 - Optimized for batch processing (256 queries per iteration) with optional GPU acceleration.
- Evaluation Framework
 - Assessed model performance using Mean Average Precision (mAP@5), Precision, Recall, and F1-Score.
 - Generated 1.2M+ labeled image pairs for supervised validation.
 - Compared retrieved neighbors against metadata-derived labels for accuracy assessment.
- Reproducibility & Code Structure
 - Organized repository into data processing, model training, evaluation, and configuration modules.
 - Due to storage constraints, large datasets were excluded, but preprocessing scripts were well-documented.

Key Learnings & Takeaways

- Scalability & Efficiency: Successfully handled combined 30GB+ datasets with batch processing and optimized memory management.
- Model Robustness: Balanced generalization and interpretability using regularization, adaptive learning, and multiple evaluation strategies.
- Search & Retrieval: Effectively combined deep learning embeddings with efficient FAISS-based vector search.
- Reproducibility: Structured code for easy reimplementing and experimentation.

This experience provided a strong foundation in classification, similarity modeling, and large-scale data processing, while reinforcing my ability to quickly adapt to new ML domains given the right resources.

Contribution Plan for Painting in a Painting

Overview

The goal of this project is to leverage artificial intelligence and multi-spectral imaging techniques to analyze paintings, extract useful information, and uncover hidden images beneath artworks. By automating and improving the analysis of paintings, this project aims to assist art historians and conservationists in understanding an artwork's history, pigment composition, and potential modifications over time.

Since the project scope is predefined, the following document outlines my implementation strategy and technical approach, which may evolve based on mentor guidance and further research.

Proposed Implementation Plan :-

1. Dataset Preparation

- Data Collection: Gather multi-spectral images of paintings from open datasets, museum archives, or research papers.
- Preprocessing: Normalize images, align multi-spectral bands, and apply data augmentation if needed.
- Metadata Integration: Extract relevant painting information, such as pigment type, restoration history, and previous analyses.

2. Feature Extraction & Analysis

- Spectral Signature Analysis: Use multi-spectral data to identify pigment types and detect areas of overpainting.
- Damage & Restoration Detection: Train a model to detect cracks, discoloration, and restoration marks.

3. Hidden Image Identification

- Baseline Model: Use supervised/unsupervised learning techniques to identify underlying patterns indicating hidden images.
- Deep Learning Approach: Train a CNN-based model (e.g., U-Net, ResNet) on multi-spectral images to distinguish painted-over regions.

Contrastive Learning: Explore self-supervised learning techniques to improve feature extraction from hidden layers.

4. (Optional) Hidden Image Reconstruction

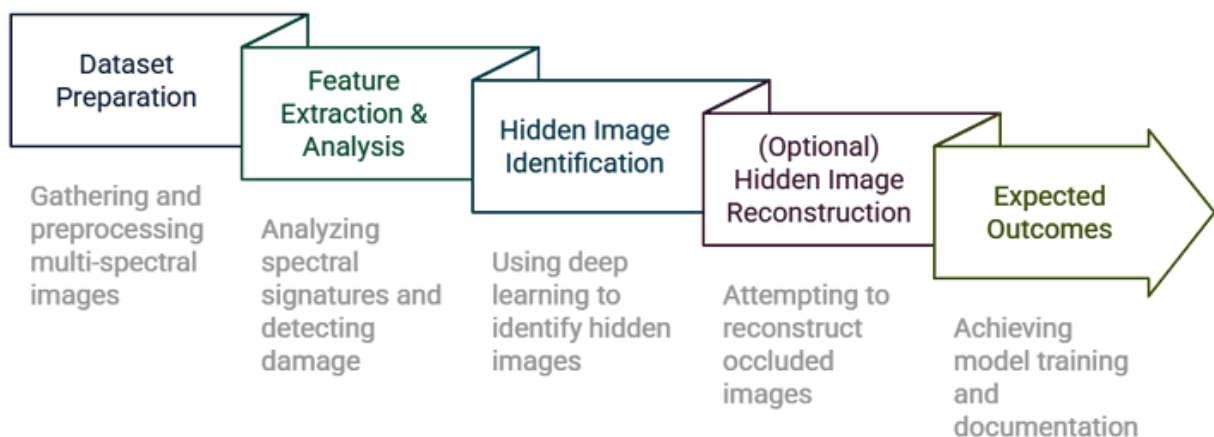
- If time permits, attempt to reconstruct occluded images using generative models (e.g., GANs, Diffusion Models).
- Compare reconstructed images with known historical references.

Expected Deliverables

- A dataset of multi-spectral images processed for model training.
- A deep learning model capable of analyzing painting properties and identifying hidden images.
- A documented pipeline for processing and analyzing multi-spectral painting data.
- (If feasible) A prototype model for reconstructing hidden images.

Tools & Technologies

- Programming Languages: Python
- Frameworks: PyTorch/TensorFlow
- Libraries: OpenCV, NumPy, SciPy, PIL, scikit-learn
- Visualization: Matplotlib, Seaborn, t-SNE for feature visualization



Flexibility & Mentor Guidance

The plan outlined above is my initial approach based on my current understanding of the problem. However, I am open to feedback and guidance from mentors to refine and improve the methodology. Given that I am actively expanding my knowledge in multi-spectral imaging and AI-based restoration techniques, I look forward to collaborating with the mentors to ensure the most effective approach is taken.

Proposed Timeline

Community Bonding Period (May 8 - June 1)

Study previous work on multispectral image analysis for paintings. Research datasets and existing techniques for detecting hidden images in paintings.

Discuss the detailed project scope and expectations with mentors.

Coding Phase 1 (June 2 - July 14)

Week 1-2: Implement data preprocessing pipeline for multispectral images.

Week 3-4: Train and evaluate a baseline model for identifying painting properties (pigments, damage, restorations).

Week 5-6: Develop an initial model for detecting hidden images.

Week 7: Optimize and validate initial results.

Midterm Evaluation (July 14 - 18)

- Submit a working implementation of the baseline painting analysis model.
- Review feedback and refine the approach.

Coding Phase 2 (July 14 - August 18)

Week 8-9: Improve hidden image detection using advanced deep learning techniques.

Week 10: Experiment with hidden image retrieval methods (if time permits).

Week 11: Conduct extensive testing and validation.

Week 12 (August 12 - 18): Buffer period for debugging, optimizations, and final refinements.

Final Submission (August 18 - 25)

- Finalize the model and results.
- Prepare documentation, reports, and final presentation.
- Conduct final evaluations with mentors.

Additional Considerations

Availability: I plan to dedicate around 35 hours per week to this project, including time set aside for ongoing learning and skill development where necessary. I am fully committed to ensuring steady progress and am open to extending my involvement as the project demands.

Semester Start: My new semester begins in August, so my availability may slightly decrease.

Buffer Period: The week 12 of the coding phase is reserved as a buffer for debugging, optimization, and handling unforeseen issues.



Previous Experience

- Art Classification Model: Built a ResNet50-based classifier for Artists, Genres, and Styles using the WikiArt dataset (25Gb). Optimized generalization with Batch Normalization, Dropout (0.75), and L2 Regularization. Used ReduceLROnPlateau for adaptive learning.
- Artwork Similarity Model: Developed a multimodal search system combining CLIP (512D) & SentenceTransformer (384D) embeddings. Used FAISS for fast nearest neighbor retrieval on 100K+ paintings. Evaluated performance with mAP@5, Precision, Recall, and F1-Score.

Why Me?

- ✓ Strong Experience in Art + AI: Successfully implemented classification & similarity models for paintings in GSoC test tasks.
- ✓ Technical Expertise: Proficient in Python, PyTorch, Deep Learning, and Computer Vision.
- ✓ Research-Oriented Approach: Focused on scalability, reproducibility, and AI-driven art restoration.



Future Work & Extensions

Beyond GSoC, this project can be extended to:

- Improve reconstruction models using GANs/Diffusion techniques.
- Expand datasets with museum-contributed multi-spectral scans.
- Optimize model deployment for real-world use by art conservationists.

Final Thoughts

This project aligns with my passion for art + AI research, and I am eager to contribute to the ArtExtract initiative.

While I have outlined a structured plan, I am open to mentor feedback and iterative refinements throughout the project.

I look forward to collaborating with the mentors and the GSoC community!

Appendix

Relevant Repositories

- Test Task Implementation: [Link](#)
- Includes code from both tasks on classification and similarity search for artworks.

References

- [WikiArt Dataset](#)
- [National Gallery of Art Open Access](#)
- [CLIP](#)
- [SentenceTransformers](#)
- [FAISS](#)